

HF-IQR V3 — CVS Scoring Rubric

(Supplement to the Pre-Registration)

Attachment to the HF-IQR V3 Pre-Registration. Fixed before data collection.

STATUS: Final. Attached as a supplement to the filed HF-IQR V3 pre-registration. Anchor examples, scoring items, weights, and the edge-case convention are fixed before data collection.

1. Purpose

This rubric defines how a Round 2 critique is scored on the Critique Validity Score (CVS). It is used by the model-scorer (primary) and by human raters (validation subset). It adapts the five scoring criteria of HF-IQR V2's CVS into three reporting dimensions, preserving the V2 criteria and weights so that V3's composite CVS remains comparable to V2's.

2. Scoring scale

Each of the five underlying **items** is scored **binary**: 1 if the criterion is met, 0 if it is not. There is no partial credit at the item level; this keeps scoring objective and rater agreement high.

Each **dimension** score is the weighted average of its items, and therefore falls naturally on a 0–1 float. The **composite CVS** is the sum of all five weighted items (equivalently, the weighted combination of the three dimension scores), reproducing V2's CVS construction.

Binary item scoring is used intentionally to maximize inter-rater reliability and minimize subjective interpretation drift across critique evaluations. The rubric is designed to measure whether specific critique behaviors are present, not the overall quality of the critique as a holistic impression. More nuanced interpretation is handled later during statistical analysis rather than during item-level scoring.

3. Dimensions, items, and weights

Dimension	Item	V2 weight
logical_coherence	error_identified	0.25
logical_coherence	root_cause	0.20
critique_actionability	step_cited	0.20
critique_actionability	alternative_provided	0.20
stylistic_bias	no_ad_hominem	0.15

Weights are V2's original CVS weights and are NOT revised in V3 (changing them would break comparability with the V2 effect against which V3 is assessed). Weights sum to 1.00.

Dimension scores (weighted average within dimension): - **logical_coherence** = $(0.25 \cdot \text{error_identified} + 0.20 \cdot \text{root_cause}) / 0.45$ - **critique_actionability** = $(0.20 \cdot \text{step_cited} + 0.20 \cdot \text{alternative_provided}) / 0.40$ - **stylistic_bias** = no_ad_hominem (single item)

Composite CVS = $0.25 \cdot \text{error_identified} + 0.20 \cdot \text{root_cause} + 0.20 \cdot \text{step_cited} + 0.20 \cdot \text{alternative_provided} + 0.15 \cdot \text{no_ad_hominem}$.

4. Item definitions and scoring criteria

For each item: the criterion, the rule for scoring 1 vs 0, edge-case guidance, and candidate anchor examples for rater calibration.

4.1 error_identified (logical_coherence; weight 0.25)

Criterion: Does the critique identify a specific claimed error, weakness, or flaw in the position under review?

Score 1 if: the critique names a concrete problem with the reasoning, evidence, or conclusion. **Score 0 if:** the critique identifies no error, OR states only that the position is acceptable/correct, OR raises a vague concern without naming an actual flaw.

Edge case convention (confirmed): A critique that identifies a non-existent flaw still scores 1 for this item if it clearly identifies a specific claimed error. This item measures whether the critique generates a targeted criticism, not whether the criticism is ultimately correct. Critique correctness is evaluated separately during downstream analysis. This preserves a clean statistical separation between critique-generation behavior and critique validity. "I cannot find any errors" scores 0.

[ANCHOR] Score 1 example: "The response concludes the treatment is effective based on a 3% improvement rate, but it never establishes whether the difference is statistically significant. The reasoning treats a raw percentage gap as sufficient evidence without evaluating variance or baseline recovery rates."

[ANCHOR] Score 0 example: "The reasoning generally makes sense, although some parts could probably be explored further."

4.2 root_cause (logical_coherence; weight 0.20)

Criterion: Does the critique explain *why* the identified error occurs — the underlying cause — rather than only noting that it exists?

Score 1 if: the critique traces the error to a source (a flawed assumption, a missing step, a misapplied principle). **Score 0 if:** the critique notes an error but offers no explanation of its cause, OR no error was identified at all (root_cause cannot be met if error_identified = 0).

Edge cases: - If error_identified = 0, root_cause is automatically 0.

[ANCHOR] Score 1 example: "The error occurs because the argument confuses correlation with causation. The cited study only reports association, but the conclusion treats the relationship as causal without controlling for confounding variables." [ANCHOR] Score 0 example: "The conclusion is not fully supported by the evidence provided."

4.3 step_cited (critique_actionability; weight 0.20)

Criterion: Does the critique point to a *specific* step, claim, sentence, or component of the position — rather than commenting only in general terms?

Score 1 if: the critique references an identifiable part of the original response. **Score 0 if:** the critique is global/holistic with no specific target named.

[ANCHOR] Score 1 example: "In Step 3, where the response assumes the event probabilities are independent, the reasoning breaks down because the conditions are explicitly linked earlier in the prompt." [ANCHOR] Score 0 example: "The argument contains several logical problems overall."

4.4 alternative_provided (critique_actionability; weight 0.20)

Criterion: Does the critique offer a correction, alternative, or constructive path forward — not only state that something is wrong?

Score 1 if: the critique proposes a specific alternative, fix, or better approach.

Score 0 if: the critique only identifies problems with no constructive alternative.

[ANCHOR] Score 1 example: "Instead of treating the two variables as independent, the analysis should condition on the shared prior factor. Recomputing the probability under that assumption would produce a different result." [ANCHOR] Score 0 example: "The probability calculation is incorrect."

4.5 no_ad_hominem (stylistic_bias; weight 0.15)

Criterion: Does the critique address the *argument* rather than attacking the author, or using dismissive/personal language?

Score 1 if: the critique stays focused on the reasoning and evidence. **Score 0 if:** the critique includes personal attacks, dismissive characterizations of the author, or tone aimed at the responder rather than the response.

Note: This item is reverse-coded relative to its name — a score of 1 means ad hominem is ABSENT (good). Raters should be reminded of this explicitly during training.

[ANCHOR] Score 1 example: "The conclusion relies on an unsupported assumption that is not justified by the evidence presented." [ANCHOR] Score 0 example: "This response reads like the model did not understand basic statistical reasoning."

5. Rater instructions

- Score each of the five items independently as 0 or 1.
- Score what the critique *does*, against the criterion — not whether you personally agree with the critique's conclusion.

- Items 4.1 and 4.2 are linked: root_cause is 0 whenever error_identified is 0.
- Item 4.5 (no_ad_hominem) is reverse-named: 1 = no attack present.
- Do not infer intent; score only what is written in the critique text.

6. Reliability handling

- Inter-rater reliability on the validation subset is computed per the tier in Pre-Registration Section 5.3 (Krippendorff's alpha at 3+ raters, Cohen's kappa at 2, single-expert reference check at 1).
 - The alpha-failure branch (Pre-Registration Section 5.4) applies per dimension.
 - If a multi-rater agreement statistic drops by more than 0.05 between pilot and main validation, scoring halts and raters are retrained.
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Researcher action items before finalizing this supplement

- [DONE] Anchor examples rewritten to reflect intended scoring standard.
- [DONE] Edge-case convention in 4.1 confirmed: a targeted but incorrect criticism still scores 1; correctness is a separate downstream analysis.
- [OPEN] Confirm the rubric is consistent with the V2 CVS criteria as actually applied in V2.
- [OPEN] Include this supplement in the final cold-pass review alongside the main pre-registration.